

## Volume 05 - Enterprise Data Mesh

# Architecture Decision Record Register

*Fifteen reusable decision records extracted and expanded from the detailed Volume 05 document.*

## ADR Index

| ADR ID       | Decision title                                                      | Status                           | Date       |
|--------------|---------------------------------------------------------------------|----------------------------------|------------|
| MESH-ADR-001 | Adopt federated data mesh instead of centralized-only data delivery | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-002 | Use domain-owned data products as the primary delivery unit         | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-003 | Use Knowledge Catalog as enterprise metadata and discovery plane    | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-004 | Use Analytics Hub for managed product sharing                       | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-005 | Use BigQuery and BigLake as standard product interfaces             | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-006 | Mandate data contracts for certified products                       | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-007 | Use policy tags, row policies, and masking for sensitive products   | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-008 | Require product-level SLOs and observability                        | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-009 | Adopt domain project factory patterns                               | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-010 | Use group-based access and avoid direct user grants                 | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-011 | Implement certification before marketplace publication              | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-012 | Treat AI-ready products as a stricter certification class           | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-013 | Use chargeback and showback at product and domain level             | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-014 | Use versioned deprecation instead of breaking product changes       | Proposed for enterprise adoption | 2026-07-11 |
| MESH-ADR-015 | Use domain stewards as certification evidence owners                | Proposed for enterprise adoption | 2026-07-11 |

# MESH-ADR-001: Adopt federated data mesh instead of centralized-only data delivery

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Centralized data teams cannot scale product delivery across hundreds of domains.

## Context

A Fortune 100 enterprise needs consistent platform controls while allowing business domains to deliver data products at their own pace. A centralized-only delivery organization becomes a queue and weakens business accountability.

## Decision drivers

- Domain scalability
- Business accountability
- Standardized controls
- Reduced delivery bottlenecks
- Regulatory traceability

## Alternatives considered

- Centralized platform and delivery team only
- Uncoordinated decentralized data marts
- Federated data mesh with central platform capabilities

## Decision

Adopt a federated data mesh with a central self-service platform and domain-owned data products.

## Rationale

The model preserves enterprise guardrails while moving product accountability and delivery closer to business knowledge.

## Tradeoffs

Requires stronger product ownership and governance automation.

## Risks

Organizational adoption, inconsistent execution, or control gaps if automation is incomplete.

## Mitigations

- Executive sponsorship
- Domain KPIs
- Automated policy gates
- Progressive maturity model
- Platform onboarding support

## Implementation implications

Create a platform product team, establish domain onboarding, deploy project and data-product factories, and pilot two domains before broader rollout.

## Cost implications

Initial platform and enablement investment increases, while long-term central delivery bottlenecks and duplicate pipelines should decrease.

## Security implications

Central security controls remain non-negotiable; domains receive delegated permissions only through approved patterns.

## Operational implications

Certified products require named support ownership, SLOs, dashboards, runbooks, and escalation paths.

## Review trigger

Annual review or material change to Google Cloud capabilities, regulation, or enterprise operating model.

# MESH-ADR-002: Use domain-owned data products as the primary delivery unit

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Datasets without product commitments are hard to trust or reuse.

## Context

The enterprise has many technically accessible datasets but inconsistent ownership, semantics, quality, and support commitments.

## Decision drivers

- Trust
- Reusability
- Accountability
- Consumer experience
- Measurable service levels

## Alternatives considered

- Dataset ownership only
- Report or dashboard ownership
- Domain-owned data product ownership

## Decision

Certified data products become the standard unit of enterprise data consumption.

## Rationale

A product definition combines data, interface, documentation, quality, access, support, and lifecycle commitments.

## Tradeoffs

Adds metadata, quality, support, and SLO obligations.

## Risks

Teams may rename datasets as products without assuming product responsibilities.

## Mitigations

- Certification checklist
- Product templates
- Product owner training
- Marketplace publication gates

## Implementation implications

Create a product descriptor, contract template, SLO template, ownership model, and certification workflow integrated with CI/CD.

## Cost implications

Product management adds operational cost but reduces duplicated discovery, reconciliation, and downstream repair work.

## Security implications

Each product carries classification, permitted uses, access groups, and policy enforcement requirements.

## Operational implications

Product owners and technical custodians jointly own incident response and lifecycle management.

## Review trigger

Annual review or material change to data-product operating standards.

# MESH-ADR-003: Use Knowledge Catalog as enterprise metadata and discovery plane

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Consumers need a single context layer for products and governance.

## Context

Metadata is fragmented across engineering tools, spreadsheets, and team-specific documentation, preventing dependable discovery and governance evidence.

## Decision drivers

- Enterprise discovery
- Metadata consistency
- Lineage
- Governance evidence
- AI-ready context

## Alternatives considered

- Spreadsheet inventory
- Third-party catalog as primary system
- Knowledge Catalog as Google Cloud-native governance plane

## Decision

Use Knowledge Catalog as the Google Cloud-native catalog and governance context layer.

## Rationale

It provides the default metadata plane for Google Cloud data assets and can anchor product discovery, classifications, glossary terms, and certification evidence.

## Tradeoffs

Requires integration and metadata-completeness discipline.

## Risks

Incomplete metadata could make the catalog untrusted.

## Mitigations

- CI/CD metadata sync
- Mandatory ownership fields
- Steward scorecards
- Certification gates

## Implementation implications

Define mandatory metadata aspects, integrate product pipelines, register products, and publish glossary and classification standards.

## Cost implications

Scanning, metadata processing, integration, and stewardship have ongoing costs that must be budgeted.

## Security implications

Catalog visibility must not expose restricted metadata or reveal sensitive asset existence to unauthorized users.

## Operational implications

Platform team operates integrations; domain stewards maintain business metadata.

## Review trigger

Annual review or significant change to Google Cloud metadata services.

# MESH-ADR-004: Use Analytics Hub for managed product sharing

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Consumers need auditable subscription patterns.

## Context

Direct dataset IAM and unmanaged copies make ownership, consumer inventory, versioning, and revocation difficult to govern.

## Decision drivers

- Auditable sharing
- Producer-consumer separation
- Discoverability
- Controlled subscription
- Reduced copying

## Alternatives considered

- Direct dataset IAM
- Manual data copies
- Analytics Hub exchanges and listings

## Decision

Publish certified products through Analytics Hub exchanges and listings where appropriate.

## Rationale

Managed listings formalize producer-consumer relationships and reduce broad direct grants.

## Tradeoffs

Requires a marketplace operating model.

## Risks

Listings may be published without adequate consumer-use controls.

## Mitigations

- Exchange types
- Publication standards
- Legal and security approval for external sharing
- Subscription recertification

## Implementation implications

Define internal, restricted, partner, and external exchange patterns with owner, steward, and security approvals.

## Cost implications

Sharing can reduce duplicate storage but may increase producer query and support costs; attribution rules are required.

## Security implications

Use approved exchanges, subscriber allowlists, row/column controls, and contractual restrictions.

## Operational implications

Marketplace operations maintain exchange taxonomy, publication workflow, and subscription reviews.



## Review trigger

Annual review or material change to Analytics Hub capabilities.

# MESH-ADR-005: Use BigQuery and BigLake as standard product interfaces

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Products must support warehouse and lakehouse patterns.

## Context

The enterprise needs high-performance SQL products and governed access to open-format data without proliferating incompatible serving technologies.

## Decision drivers

- Standard interfaces
- SQL performance
- Open formats
- Fine-grained access
- Interoperability

## Alternatives considered

- BigQuery only
- Object storage only
- Mixed unmanaged interfaces
- BigQuery plus BigLake standards

## Decision

Use BigQuery for curated analytical products and BigLake for governed open-format products.

## Rationale

The combination supports warehouse and lakehouse consumption while retaining a common Google Cloud governance and access model.

## Tradeoffs

Requires file-layout and interface-selection standards.

## Risks

Domains may expose raw storage as a product without sufficient curation.

## Mitigations

- Interface decision matrix
- File compaction standards
- Certification tests
- Query performance thresholds

## Implementation implications

Publish standard Terraform modules, dataset policies, BigLake connection patterns, and file-format guidelines.

## Cost implications

BigQuery compute and storage, BigLake metadata/query costs, and cross-region transfer must be measured by product.

## Security implications

Use dataset IAM, policy tags, authorized views, and controlled connections rather than direct bucket grants.

## Operational implications

Product teams monitor freshness, query performance, cost, and file health.

## Review trigger

Annual review or material change to BigQuery or BigLake architecture.

# MESH-ADR-006: Mandate data contracts for certified products

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Consumers need predictable schema, semantics, and freshness.

## Context

Unmanaged schema changes and undocumented semantics create downstream incidents and hidden coupling.

## Decision drivers

- Interface stability
- Consumer trust
- Change management
- Quality
- Automation

## Alternatives considered

- Informal documentation
- Schema-only contracts
- Full data-product contracts

## Decision

Full data contracts are mandatory for product certification.

## Rationale

Contracts provide a testable agreement for schema, semantics, SLOs, classifications, ownership, and change policy.

## Tradeoffs

Increases initial product effort.

## Risks

Contracts may drift from deployed schemas.

## Mitigations

- CI validation
- Schema comparison
- Contract repository
- Release gates

## Implementation implications

Provide YAML/JSON contract templates, schema registry integration, compatibility checks, and release approval workflow.

## Cost implications

Additional engineering effort is offset by lower incident and rework costs.

## Security implications

Contracts include classification, permitted use, retention, and privacy constraints.

## Operational implications

Contract violations generate alerts and can block publication.

## Review trigger

Annual review or material change to product contract standards.

# MESH-ADR-007: Use policy tags, row policies, and masking for sensitive products

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Sensitive products need fine-grained controls.

## Context

Dataset-level access is too coarse for products containing mixed sensitivity, geography, or legal-use restrictions.

## Decision drivers

- Least privilege
- Privacy
- Regulatory compliance
- Reusable products
- Reduced copying

## Alternatives considered

- Dataset-only IAM
- Separate masked copies
- Fine-grained policies with authorized interfaces

## Decision

Use policy tags, row access policies, dynamic masking, and authorized interfaces for sensitive products.

## Rationale

Fine-grained controls allow one governed product to support multiple authorized consumer populations.

## Tradeoffs

Requires taxonomy and entitlement governance.

## Risks

Misclassification or policy misconfiguration may expose sensitive data.

## Mitigations

- Automated classification
- Schema validation
- Policy tests
- Access reviews
- Security approval

## Implementation implications

Create enterprise taxonomies, mapping rules, test personas, and automated policy deployment.

## Cost implications

Policy management adds governance overhead but reduces storage duplication and copy management.

## Security implications

This is a primary control decision; controls must be validated with positive and negative access tests.

## Operational implications

Security and domain teams jointly own access incidents and recertification.

## Review trigger

Annual review or regulatory/control-model change.

# MESH-ADR-008: Require product-level SLOs and observability

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Product trust depends on measurable operations.

## Context

Platform availability alone does not prove that a particular product is fresh, complete, accurate, or available to consumers.

## Decision drivers

- Trust
- Operational accountability
- Consumer transparency
- Incident management
- Capacity planning

## Alternatives considered

- Best-effort support
- Platform-only SLOs
- Product-specific SLOs

## Decision

Each certified product has freshness, quality, availability, and support SLOs.

## Rationale

Product-specific service objectives make ownership enforceable and allow consumers to select products based on reliability.

## Tradeoffs

Requires monitoring cost and operational ownership.

## Risks

Teams may define weak or non-actionable SLOs.

## Mitigations

- Standard SLI catalogue
- Tier-based defaults
- Generated dashboards
- Periodic SLO review

## Implementation implications

Deploy standard metrics, dashboards, alert policies, product scorecards, and error-budget reporting.

## Cost implications

Monitoring, logging, and support costs must be allocated to the product.

## Security implications

Security-related SLOs include access-review completion and unauthorized-access alert response.

## Operational implications

Product teams own alerts, runbooks, on-call routing, and post-incident review.



## Review trigger

Quarterly SLO review and annual architecture review.

# MESH-ADR-009: Adopt domain project factory patterns

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Domain autonomy needs repeatable infrastructure.

## Context

Manual project provisioning and shared monolithic projects create inconsistent controls and slow onboarding.

## Decision drivers

- Repeatability
- Isolation
- Speed
- Compliance
- Scalability

## Alternatives considered

- Manual projects
- Single central shared project
- Terraform-based domain project factory

## Decision

Provision standardized domain projects, datasets, identities, budgets, and monitoring through Terraform.

## Rationale

A project factory turns the landing-zone and mesh guardrails into a repeatable self-service capability.

## Tradeoffs

Requires initial and ongoing module investment.

## Risks

Teams may bypass the factory for urgent delivery.

## Mitigations

- Policy enforcement
- Fast onboarding SLA
- Exception workflow
- Drift detection

## Implementation implications

Create golden Terraform modules, request workflow, validation pipeline, and lifecycle management.

## Cost implications

Factory investment reduces repeated engineering and audit effort over time.

## Security implications

Factory modules enforce IAM, logging, KMS, perimeter, and network defaults.

## Operational implications

Platform team owns modules and upgrades; domains own resources created within their approved boundary.

## Review trigger

Semiannual module and architecture review.

# MESH-ADR-010: Use group-based access and avoid direct user grants

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Auditability and lifecycle management require group-based control.

## Context

Direct user grants are difficult to review, revoke, and associate with enterprise roles or consumer teams.

## Decision drivers

- Least privilege
- Joiner-mover-leaver control
- Auditability
- Scalability
- Delegated administration

## Alternatives considered

- Direct user grants
- Group-based grants
- Dynamic entitlement service

## Decision

Use enterprise groups and product-specific IAM bindings by default; prohibit direct user grants except approved break-glass cases.

## Rationale

Groups align access with roles and teams and enable periodic recertification.

## Tradeoffs

Group lifecycle must be actively managed.

## Risks

Stale groups can retain excessive privileges.

## Mitigations

- Access expiry
- Owner recertification
- Automated membership review
- Break-glass monitoring

## Implementation implications

Create naming standards, access request workflows, product consumer groups, and recertification automation.

## Cost implications

Identity administration cost increases modestly but audit and remediation effort decreases.

## Security implications

No direct grants in production except logged emergency access with automatic expiry.

## Operational implications

Identity team owns group lifecycle; domain owners approve product entitlements.

## Review trigger

Quarterly access-control review.

# MESH-ADR-011: Implement certification before marketplace publication

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

A marketplace without trust signals creates risk.

## Context

Publishing unverified assets encourages reuse of incomplete, undocumented, or insecure datasets.

## Decision drivers

- Consumer trust
- Security
- Quality
- Discoverability
- Lifecycle control

## Alternatives considered

- Open publication
- Manual review only
- Automated certification gate with risk-based approval

## Decision

Products must pass defined certification gates before enterprise marketplace publication.

## Rationale

Certification establishes minimum metadata, quality, security, ownership, SLO, and lifecycle standards.

## Tradeoffs

Can slow early adoption if gates are too rigid.

## Risks

Manual certification may become a bottleneck.

## Mitigations

- Automated evidence collection
- Risk tiers
- Pilot maturity exceptions
- SLA for approvals

## Implementation implications

Define certification classes, automated checks, approval workflow, badges, and renewal criteria.

## Cost implications

Certification adds governance cost but reduces downstream quality and security incidents.

## Security implications

Restricted and external products require enhanced security and legal approval.

## Operational implications

Certification expires or is suspended when quality, SLO, ownership, or security evidence fails.

## Review trigger

Annual certification-framework review.

# MESH-ADR-012: Treat AI-ready products as a stricter certification class

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

AI use increases privacy, provenance, and misuse risk.

## Context

Data used for training, grounding, embeddings, and agents can amplify quality defects and expose sensitive information through indirect outputs.

## Decision drivers

- Responsible AI
- Privacy
- Provenance
- Grounding quality
- Permitted use

## Alternatives considered

- Allow all certified products for AI
- Block enterprise AI use
- Create a stricter AI-ready certification class

## Decision

Create an AI-ready certification class with explicit permitted uses, provenance, quality, privacy, and guardrail requirements.

## Rationale

AI-ready certification provides stronger assurances without blocking all governed AI use.

## Tradeoffs

Requires additional governance and technical evidence.

## Risks

Teams may bypass certification for experimentation.

## Mitigations

- Controlled sandboxes
- AI review checklist
- Model/data access controls
- Audit and egress monitoring

## Implementation implications

Add AI metadata, allowed-use fields, PII checks, provenance, evaluation datasets, and retrieval-quality requirements.

## Cost implications

Additional profiling, evaluation, and monitoring costs apply to AI-ready products.

## Security implications

Products require prompt-injection, data-exfiltration, privacy, and content-safety assessments where applicable.

## Operational implications

AI-ready status is periodically revalidated as data, models, and regulations change.



## Review trigger

At least semiannual review due to rapid AI capability and policy change.

# MESH-ADR-013: Use chargeback and showback at product and domain level

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Cost transparency is required for mesh sustainability.

## Context

Shared platform and query costs can become invisible, while domains have limited incentives to optimize products they publish or consume.

## Decision drivers

- Cost accountability
- Optimization
- Funding
- Product economics
- Capacity planning

## Alternatives considered

- Project-only cost reporting
- Domain-level showback
- Product-level showback and targeted chargeback

## Decision

Use mandatory product and domain labels with billing export for product-level showback and targeted chargeback.

## Rationale

Product economics help owners balance service quality, demand, and cost while retaining central platform transparency.

## Tradeoffs

Attribution is imperfect for shared workloads.

## Risks

Incorrect labels or shared compute can distort reports.

## Mitigations

- Mandatory labels
- Query job tagging
- Reservation mapping
- Allocation rules
- FinOps review

## Implementation implications

Define cost taxonomy, export billing data, publish domain dashboards, and phase from showback to selective chargeback.

## Cost implications

FinOps tooling and governance add overhead but enable measurable savings.

## Security implications

Billing views must not expose sensitive resource metadata beyond authorized stakeholders.

## Operational implications

FinOps, platform, and domain owners review cost, unit metrics, and optimization actions monthly.

## Review trigger

Quarterly cost-allocation review.

# MESH-ADR-014: Use versioned deprecation instead of breaking product changes

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Consumers need stable interfaces.

## Context

Breaking schema or semantic changes can disrupt many downstream consumers and create unplanned enterprise incidents.

## Decision drivers

- Consumer stability
- Change transparency
- Backward compatibility
- Lifecycle control
- Operational safety

## Alternatives considered

- Immediate breaking changes
- Retain all versions indefinitely
- Versioned releases with planned deprecation

## Decision

Breaking changes require a major product version, migration window, usage monitoring, and controlled retirement.

## Rationale

Versioned deprecation balances consumer stability with the need to evolve products.

## Tradeoffs

Multiple versions must be maintained temporarily.

## Risks

Old versions may persist because consumers fail to migrate.

## Mitigations

- Usage monitoring
- Retirement deadlines
- Escalation
- Migration tooling
- Executive exception approval

## Implementation implications

Use semantic versioning, compatibility tests, deprecation metadata, consumer notifications, and retirement gates.

## Cost implications

Temporary parallel versions increase cost; retirement discipline is required.

## Security implications

Critical security fixes can override normal windows with documented emergency governance.

## Operational implications

Product owners maintain migration plans and monitor remaining consumers.

## Review trigger

Review with each major version and annually.

# MESH-ADR-015: Use domain stewards as certification evidence owners

|                     |                                  |
|---------------------|----------------------------------|
| Status              | Proposed for enterprise adoption |
| Decision date       | 2026-07-11                       |
| Architecture domain | Enterprise Data Mesh             |

## Problem

Business meaning cannot be delegated entirely to engineers.

## Context

Technical metadata alone does not prove that a product has correct business definitions, quality interpretation, permitted use, or regulatory meaning.

## Decision drivers

- Business accountability
- Semantic consistency
- Quality interpretation
- Regulatory evidence
- Consumer trust

## Alternatives considered

- Engineer-only review
- Central governance-only review
- Domain steward ownership of business certification evidence

## Decision

Domain data stewards own business metadata, glossary alignment, quality interpretation, and certification evidence.

## Rationale

Stewards are closest to business meaning and can validate whether technical results are fit for intended use.

## Tradeoffs

Requires dedicated staffing and enablement.

## Risks

Underfunded stewardship can delay certification or reduce quality.

## Mitigations

- Steward staffing model
- Training
- Workflow automation
- Delegation rules
- Performance metrics

## Implementation implications

Assign stewards per product/domain, define RACI, automate evidence collection, and create approval queues.

## Cost implications

Stewardship is an explicit operating cost and should be included in domain funding.

## Security implications

Stewards validate classification and permitted-use metadata but do not replace security approval.

## Operational implications

Stewards participate in quality incidents, glossary changes, and periodic recertification.

## Review trigger

Annual operating-model review or material governance change.